

1. NAME OF THE MEDICINAL PRODUCT

LOCOA 40mg transdermal patch

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 10 × 14 cm patch contains 40mg of esflurbiprofen.
For a full list of excipients, see Section 6.1.

3. PHARMACEUTICAL FORM

Transdermal Patch

The patch consists of a colorless to light yellow transparent adhesive plaster base and has characteristic aroma, in which the adhesive plaster base is spread on a backing layer, and the surface of the adhesive plaster base is covered with a liner.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Relief of pain and inflammation associated with osteoarthritis

4.2 Posology and method of administration

Posology

LOCOA should be applied to the affected area once daily in adults. Do not use more than 2 patches at once.

Do not use more than 2 patches per day, since LOCOA is a topical patch with improved transdermal absorption and enhanced penetration of esflurbiprofen, the active ingredient, into the affected tissue. Also, the safety of LOCOA has been examined for up to 2 patches per day in clinical studies. The use of LOCOA with any other anti-inflammatory agent with expected systemic effects should be avoided wherever possible. If such use is deemed necessary, it should be limited to a minimal level and the condition of the patient should be carefully monitored.

Elderly

Since elderly patients are more susceptible to adverse reactions, their condition should be carefully monitored during use of LOCOA.

Paediatric population

Safety has not been established in low-birth-weight babies, neonates, nursing infants, infants and children. [Clinical studies have only been performed in adults and there is no experience in the use for the above patients.]

Method of administration

LOCOA is administered topically by attaching the patch to the skin of the affected area.

LOCOA cannot be applied to the skin wounds, open injuries, mucous membrane or to the area of eczema or rash. If 2 patches are used at a time, the entire sticky side needs to be in direct contact with the skin. The patches should not be placed overlap or on top of another. LOCOA should be applied to clean and dry skin. Careful administration is required with attention to the skin condition of

the application site. LOCOA should be detached slowly and carefully to prevent any damage to the skin.

4.3 Contraindications

- Patients with peptic ulcer (See section 4.4.)
- Patients with severe blood abnormality.
- Patients with severe liver disorder.
- Patients with severe renal disorder.
- Patients with severe cardiac dysfunction.
- Patients with severe hypertension.
- Patients with a history of hypersensitivity to the components of LOCOA or flurbiprofen.
- Patients with current or previous aspirin-induced asthma (asthmatic attacks induced by nonsteroidal anti-inflammatory analgesics and other relevant drugs.) (See section 4.8.)
- Patients on enoxacin hydrate, lomefloxacin, norfloxacin, or prulifloxacin (See section 4.5.)
- Women in the third trimester of pregnancy (See section 4.6.)

4.4 Special warnings and precautions for use

General

- A thorough health interview should be performed to predict hypersensitivity reactions.
- For long term use, periodic laboratory tests (urine analysis, blood test, liver function test, etc.) should be performed. If any abnormality is observed, appropriate measures such as dose interruption should be taken (See section 4.8.)
- Patients should be monitored closely for adverse reactions. Symptoms such as an excessive decrease in body temperature, collapse and coldness of limbs may occur and hence, this drug should be administered with great care while monitoring their condition in the elderly with hyperthermia or patients with debilitating diseases.
- LOCOA must not be used with enoxacin hydrate, lomefloxacin, norfloxacin or prulifloxacin since convulsions have been reported in a few patients after receiving flurbiprofen axetil with these new quinolone antibiotics. The use of LOCOA with other new quinolone antibiotics should be avoided wherever possible.
- If cutaneous symptoms are observed with the use of LOCOA, appropriate measures such as drug interruption or discontinuation should be taken according to symptoms.
- Temporary infertility has been reported in women after long term use of nonsteroidal anti-inflammatory analgesics.

Careful administration

LOCOA should be administered with care in the following patients.

Patients with peptic ulcer induced by long term use of nonsteroidal anti-inflammatory analgesics who need long term use of LOCOA and are being treated with agents such as misoprostol

Since some patients may have peptic ulcer resistant to agents such as misoprostol, adequate monitoring and careful administration is required, when the use of LOCOA is continued.

Patients with a history of peptic ulcer

Gastric mucosal barrier decreased by inhibition of prostaglandin synthesis may cause a relapse of peptic ulcer.

Patients with current or previous blood abnormality

Blood abnormality may be aggravated or recur by development of blood disorder.

Patients with bleeding tendency

Platelet function may decrease, increasing bleeding tendency.

Patients with current or previous liver disorder

Liver disorder may be aggravated or recur by hepatic function abnormality.

Patients with current or previous renal disorder, or reduced renal blood flow

Renal disorder may be aggravated or recur, or may be induced by reduced renal blood flow via inhibition of prostaglandin synthesis.

Patients with abnormal cardiac function

Water and sodium retention induced by inhibition of prostaglandin synthesis may aggravate cardiac dysfunction.

Patients with hypertension

Water and sodium retention induced by inhibition of prostaglandin synthesis may further increase blood pressure.

Patients with a history of hypersensitivity to esflurbiprofen or flurbiprofen, or any of the other excipients

For the full list of excipients, see section 6.1.

Patients with bronchial asthma

Make sure that the patient does not have aspirin-induced asthma. Some patients with bronchial asthma may actually have aspirin-induced asthma, with which asthmatic attacks may be induced (See section 4.8.)

Patients with ulcerative colitis

Ulcerative colitis may be aggravated.

Patients with Crohn's disease

Symptoms aggravated by other nonsteroidal anti-inflammatory analgesics have been reported.

Use in the elderly

Elderly patients are more susceptible to adverse reactions. Care should be taken in administering LOCOA to elderly patients, for example, limiting the use to a minimal level, while special attention should be paid to possible adverse reactions.

Paediatric use

Safety has not been established in low-birth-weight babies, neonates, babies, infants and children. [There is no experience in the use for the above patients.]

Risk of GI ulceration, bleeding and perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Use this medication by prescription only.

4.5 Interaction with other medicinal products and other forms of interactionContraindications for concomitant administration***Enoxacin hydrate, lomefloxacin, norfloxacin, prulifloxacin***

Concomitant use may induce convulsions due to enhancing GABA inhibition by new quinolone antibiotics (See section 4.3.)

Precautions for concomitant administration***New quinolone antibiotics (ofloxacin, etc.) [Note that enoxacin hydrate, lomefloxacin, norfloxacin and prulifloxacin are contraindicated for concomitant administration]***

Concomitant use may induce convulsions due to enhancing GABA inhibition by new quinolone antibiotics.

Coumarin anticoagulants (warfarin)

Care such as dose adjustment should be taken to avoid possible increase in effect of coumarin anticoagulants (warfarin). Eslurbiprofen may compete with warfarin in plasma protein binding, increasing free warfarin.

Methotrexate

Care such as dose adjustment should be taken to avoid possible poisoning symptoms (e.g., anaemia and thrombocytopenia) due to increase in the effect of methotrexate.

Inhibition of prostaglandin synthesis by esflurbiprofen may reduce renal blood flow, resulting in decreased renal excretion and thus increased blood levels of methotrexate.

Lithium products (lithium carbonate)

Careful measures such as adequate monitoring of lithium blood levels should be taken during the use of lithium products, since concomitant use may increase blood levels of lithium, which may cause lithium poisoning symptoms. Inhibition of prostaglandin synthesis by esflurbiprofen may reduce renal excretion of sodium and lithium clearance, thus increasing lithium blood levels.

Thiazide diuretics (hydrochlorothiazide, etc.) and loop diuretics (furosemide, etc.)

The effect of these diuretics may decrease. Inhibition of prostaglandin synthesis by esflurbiprofen may produce water and salt retention in the body.

Corticosteroids (methylprednisolone, etc.)

Concomitant use may mutually increase gastrointestinal reactions (peptic ulcer, gastrointestinal bleeding, etc.). Concomitant use may increase gastrointestinal reactions to both drugs.

Pharmacokinetic interaction with CYP2C9 inhibitors (fluconazole, etc.)

Eslurbiprofen is mainly metabolized by hepatic enzyme CYP2C9. Eslurbiprofen blood levels may increase, since metabolism of esflurbiprofen is inhibited by concomitant use of CYP2C9 inhibitors.

4.6 Fertility, pregnancy and lactation

- LOCOA cannot be used in women in the third trimester of pregnancy. A study in rats in late pregnancy showed deaths of dams, delayed delivery, reduced fertility rates, and increased stillbirths with less than the plasma esflurbiprofen level (AUC) equivalent to that yielded by the use of 2 patches of LOCOA in humans (See section 4.3 and 5.3.)
- LOCOA should be administered to pregnant women (excluding those in their third trimester) or women who may be pregnant only when the therapeutic benefits are considered to outweigh the risks. Safety has not been established in pregnant women. If such administration is deemed necessary, caution should be exercised such as limiting to the minimum effective use and monitoring for amniotic fluid and suspected findings of ductus arteriosus constriction as necessary, considering gestational weeks (around 20 weeks gestation or later in pregnancy) and administration periods. Renal impairment and decreased urine output in foetuses as well as accompanying oligohydramnios have been reported following use of cyclooxygenase inhibitors (oral dosage form or suppository) in pregnant women. In addition, ductus arteriosus constriction has been reported following use of cyclooxygenase inhibitors (products with expected systemic effects) in women in the second trimester.
- When LOCOA is used to lactating women, continuation or discontinuation of lactating should be determined based on consideration of the potential therapeutic benefits and the benefits of lactating. [An animal study (rats) showed excretion of esflurbiprofen into the milk and restricted increase in body weight in offspring of dams with approximately 3 times the plasma level (AUC) yielded by the use of 2 patches of LOCOA in humans (See section 5.3.)]
- Constriction of the fetal ductus arteriosus has been reported in women in the third trimester of pregnancy using other topical nonsteroidal anti-inflammatory analgesics.
- Temporary infertility has been reported in women after long term use of nonsteroidal anti-inflammatory analgesics.

4.7 Effects on ability to drive and use machines

It is considered that application of LOCOA at recommended doses has no influence on the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

In the clinical studies performed in a total of 1,391 patients, 415 adverse reactions were reported in 269 patients (19.3%), including application site dermatitis in 111 patients (8.0%), application site erythema in 44 patients (3.2%), and application site eczema in 32 patients (2.3%).

Tabulated list of adverse reactions reported in the clinical studies and from post-marketing experience

The following adverse reactions have been reported in all the clinical trials and from post-marketing experience with LOCOA. Adverse reactions listed below are classified according to frequency and system organ class (SOC).

Table 1. Adverse reactions in all the clinical trials and from post-marketing experience with LOCOA

	Common (≥ 1/100 to < 1/10)	Uncommon (≥ 1/1,000 to < 1/100)	Incidence unknown
Application site disorders	Dermatitis, Erythema, Pruritus, Eczema, Rash	Bruise, Irritation, Pigmentation	Oedema, Discolouration, Pain, Warmth
Nervous system disorders		Dizziness	Headache
Gastrointestinal disorders		Abdominal discomfort, Gastritis, Peptic ulcer, Abdominal pain, Nausea, Vomiting, Stomatitis	Constipation, Diarrhoea, Decreased appetite, Faeces discoloured
Hypersensitivity		Rash	Angioedema (face, eyelid, etc.), Eczema, Erythema, Urticaria, Flushing, Pruritus
Investigations	Blood urea increased	Blood creatinine increased, AST increased, ALT increased, Blood urine present, Blood bilirubin increased, Blood lactate dehydrogenase increased, Glucose urine present, Protein urine present	Blood pressure increased
Others		Palpitations	Oedema peripheral

Clinically significant adverse reactions

Shock, anaphylaxis

Since shock and anaphylaxis (with an unknown incidence for both) may occur, the condition of patients should be carefully monitored. If symptoms such as chest discomfort, chills, cold sweat, dyspnea, numbness in extremities, decreased blood pressure, angioedema, and urticaria are observed, the use of LOCOA should be discontinued while taking appropriate actions.

Acute kidney injury, nephrotic syndrome

Since acute kidney injury may occur, this medicine may cause serious renal disorders including acute kidney injury and nephrotic syndrome (with an unknown incidence for all). So the condition of patients should be carefully monitored, for example, with periodic tests. If signs such as oliguria, hematuria, protein urine, increases in BUN and blood creatinine, hyperkalaemia, and hypoalbuminaemia are observed, appropriate actions should be taken such as discontinued use of LOCOA (See section 4.4.)

Gastrointestinal bleeding

Since gastrointestinal bleeding (with an unknown incidence) may occur, the condition of patients should be carefully monitored. If any abnormality is observed, the use of LOCOA should be discontinued while taking appropriate treatment.

Aplastic anaemia

Since aplastic anaemia (with an unknown incidence) may occur, the condition of patients should be carefully monitored. If any abnormality is observed, appropriate actions should be taken such as discontinued use of LOCOA.

Induced asthma attacks (aspirin-induced asthma)

Asthma attacks (with an unknown incidence) may occur. If initial symptoms such as dry rales, wheezing, and dyspnea are observed, the use of LOCOA should be discontinued.

Toxic epidermal necrolysis (TEN), Stevens-Johnson syndrome, exfoliative dermatitis

Since toxic epidermal necrolysis, Stevens-Johnson syndrome, and exfoliative dermatitis (with an unknown incidence for both) may occur. So the condition of patients should be carefully monitored. If any abnormality is observed, the use of LOCOA should be discontinued while taking appropriate treatment.

Impaired consciousness, convulsions with impaired consciousness

Since impaired consciousness and convulsions with impaired consciousness have been reported as a rare incidence (< 0.1%) with the use of flurbiprofen axetil [injections], the condition of patients should be carefully monitored. If any abnormality is observed, the use of LOCOA should be discontinued while taking appropriate treatment.

Myocardial infarction, cerebrovascular disorder

Since cardiovascular thromboembolic events such as myocardial infarction and cerebrovascular disorder (with an unknown incidence for both) may occur, the condition of patients should be carefully monitored. If any abnormality is observed, appropriate actions should be taken such as discontinued use of LOCOA.

4.9 OverdoseSymptoms

Symptoms of overdose may include headache, nausea, vomiting, epigastric pain, gastrointestinal bleeding, rarely diarrhoea, disorientation, excitation, coma, drowsiness, dizziness, tinnitus, fainting and occasionally convulsions. In cases of significant poisoning, acute renal failure and liver damage are possible.

Therapeutic measures

Patients should be treated symptomatically as required.

Patients should be advised to immediately remove the patches and wash the skin with water, then contact the doctor, pharmacist, or nearest hospital immediately.

5. PHARMACOLOGICAL PROPERTIES**5.1 Pharmacodynamic properties**Mechanism of action

Esflurbiprofen is an optical isomer (S isomer) of flurbiprofen, racemate.

An *in vitro* study demonstrated the inhibitory action of esflurbiprofen on the cyclooxygenase activity, which is probably the main source of anti-inflammatory/analgesic effects.

Pharmacodynamic effects***Analgesic effect***

The analgesic effect was demonstrated in all pain models: a dog model of urate-induced knee joint arthritic pain, and rat models of carrageenan-induced inflammatory pain, silver nitrate-induced arthritic pain, and adjuvant-induced arthritic pain.

Anti-inflammatory effect

The anti-inflammatory effect was demonstrated in all acute inflammation models: rat models of carrageenan-induced footpad inflammation, traumatic oedema, and adjuvant-induced arthritis.

Clinical efficacy and safety

A randomized double-blind, placebo-controlled, parallel-group study (phase II dose-finding study) and a randomized open-label* parallel-group study with flurbiprofen patches as the control (phase III study) were performed in patients with knee osteoarthritis. The following tables show changes from baseline in VAS scores (pain felt on rising from the chair) in patients receiving esflurbiprofen 40mg, base, or flurbiprofen patches for 2 weeks.

*The study was performed with well managed information on treatment groups for study subjects and investigators.

Table 2. Phase II dose-finding study

	10 mg	20 mg	40 mg	Base
Baseline	57.8 ± 12.3 (121)	56.0 ± 12.5 (127)	57.0 ± 12.4 (134)	58.4 ± 13.5 (126)
Final evaluation	26.1 ± 17.5 (121)	24.5 ± 17.6 (127)	21.5 ± 16.7 (134)	28.4 ± 18.9 (126)
Change	-31.7 ± 17.1 (121)	-31.5 ± 16.1 (127)	-35.5 ± 17.1 (134)	-30.1 ± 18.8 (126)
Between-group difference ^{a)} [95% confidence interval] ^{a)} p-value ^{a),b)}	-1.9 [-6.0, 2.2]	-2.5 [-6.5, 1.5] p=0.112	-6.1 [-10.1, -2.1] p=0.001	-

Mean (mm) ± SD (No. of subjects)

a) An analysis of covariance model with explanatory variables of baseline and treatment group (one-sided significance level 0.025)

b) The fixed order approach was used to take multiplicity of the test into account in paired comparisons of the esflurbiprofen 40, 20, and 10 mg groups and the base group.

Table 3. Phase III study

	Esflurbiprofen 40 mg	Flurbiprofen patch 40mg x 2
Baseline	59.5 ± 12.7 (315)	59.3 ± 12.5 (317)
Final evaluation	18.5 ± 15.3 (315)	28.8 ± 18.1 (317)
Change	-41.0 ± 15.5 (315)	-30.5 ± 15.9 (317)
Between-group difference ^{a)} [95% confidence interval] ^{a)} p-value ^{a)}	-10.4 [-12.7, -8.0] p<0.001	-

Mean (mm) ± SD (No. of subjects)

a) An analysis of covariance model with explanatory variables of baseline and treatment group (one-sided significance level 0.025)

5.2 Pharmacokinetic properties

Plasma levels

Single dose

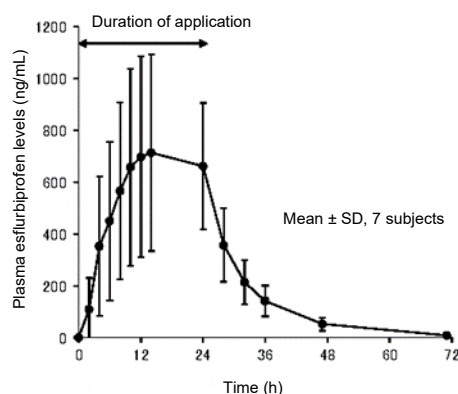
The pharmacokinetic parameters and changes in plasma concentrations in healthy adults (Japanese) receiving a single 24-hour application of 40mg of esflurbiprofen are shown in Table 4 and Figure 1. Based on the residual drug amount in the patch, the transdermal absorption was calculated to be 48.34 ± 16.70%.

Table 4. Pharmacokinetic parameters of esflurbiprofen after a single administration

Dose (No. of subjects)	C _{max} (ng/mL)	t _{max} (h)	t _{1/2} (h)	AUC _{0-∞} (ng·h/mL)
40 mg (7 subjects)	751 ± 360	17.7 ± 5.94	8.60 ± 0.615	19000 ± 9390

Mean ± SD

Figure 1. Plasma concentration-time profile of esflurbiprofen after a single 24-hour application of 40mg of esflurbiprofen in human



Multiple doses

The pharmacokinetic parameters in healthy adults (Japanese) receiving multiple 23-hour applications of once daily doses of 80 mg of esflurbiprofen for 7 days are shown in Table 5. Based on the residual drug amount in the patch, the transdermal absorption was calculated to be $73.24 \pm 11.58\%$.

Table 5. Pharmacokinetic parameters of esflurbiprofen after multiple administration

Dose (No. of subjects)	Study day	C _{max} (ng/mL)	t _{max} (h)	t _{1/2} (h)	AUC _{0-23h} (ng·h/mL)
80 mg (6 subjects)	Day 1	1360 ± 551	10.3 ± 1.51	-	23500 ± 8530
	Day 7	2710 ± 669	6.67 ± 2.07	8.13 ± 0.503	47000 ± 10100

Mean ± SD, -: No data

Distribution

Tissue penetration

In patients (Japanese) with knee osteoarthritis scheduled to undergo knee replacement arthroplasty, the levels of esflurbiprofen in the synovia, synovial fluid, and plasma after a single 12-hour application of 20 mg of esflurbiprofen were 14.8, 32.7, and 34.5 times higher, respectively, than those after the use of 40 mg flurbiprofen patch.

Protein binding

An *in vitro* study showed that 99.95% of esflurbiprofen was bound to human plasma protein, probably to albumin as the main binding protein.

Metabolism

Esflurbiprofen mainly undergoes oxidative metabolism by CYP2C9. A study investigating the effect of CYP2C9 polymorphism using human liver microsomes showed the 4'-hydroxylation activity (CL_{int}) in poor metabolizers (PMs) (with genotype CYP2C9*3/*3) was 1/69 of that in extensive metabolizers (EMs) (with genotype CYP2C9*1/*1).

Excretion

In healthy adults (Japanese) receiving a single 24-hour application of 80 mg of esflurbiprofen, little of the dose (0.253%) was excreted as unchanged compound in the urine through 72 hours after the start of application.

Major metabolites were glucuronate or sulfate conjugates of the 4'-hydroxy compound. Other metabolites were free glucuronate conjugates, 4'-hydroxy compounds, and glucuronate conjugates of the 3'-hydroxy-4'-methoxy compound.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, fertility and early development, and embryo-fetal development.

In the pre- and postnatal development study using rats in late pregnancy, maternal animal death at ≥ 0.1 mg/kg, a tendency of an increased number and percentage of stillborn pups at 0.1, 0.3, and 3 mg/kg, delayed delivery at 0.3 mg/kg, reduced live birth index at ≥ 0.3 mg/kg, and incomplete delivery and prolonged delivery at ≥ 1 mg/kg were indicated. Since the AUC_{0-24h} value of esflurbiprofen on Day 17 of gestation at 1 mg/kg was comparable or less than the exposure after 40 or 80 mg of esflurbiprofen was administered to humans, the AUC_{0-24h} values at 0.1 and 0.3 mg/kg appeared to be lower than that in humans.

After esflurbiprofen was administered to rats in the lactation period, suppression of postnatal body weight gain and delayed onset of early behavior were seen at 3 mg/kg. The AUC_{0-24h} value of esflurbiprofen on Day 17 of gestation at 3 mg/kg were 5.77 and 2.55 times higher than the exposure after 40 or 80 mg of esflurbiprofen was administered to humans.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Mentha oil
Liquid paraffin
Styrene-isoprene-styrene block copolymer
Polyisobutylene
Ester gum HG
Propylene glycol dicaprylate
2-Mercaptobenzimidazole
Dibutylhydroxytoluene

6.2 Shelf life

2 years

6.3 Special precautions for storage

Store away from direct sunlight, heat and moisture. Store below 25°C

6.4 Nature and contents of container

70 patches (7 patches × 10 aluminium foil laminated pouches) per box

6.5 Special precautions for disposal

LOCOA should not be disposed via wastewater or household waste.

7. MANUFACTURER/PRODUCT REGISTRATION HOLDER

Under Authority of:
Taisho Pharmaceutical Co., Ltd.
3-24-1 Takada, Toshima-ku, Tokyo, Japan

Manufacturer:
TOKUHON Corporation Miyashiro Factory
1010, Aza Yamazaki, Miyashiro-machi, Minamisaitama-gun, Saitama, Japan

Product Registration Holder:
HOE Pharmaceuticals Sdn. Bhd.
Lot 10, Jalan Sultan Mohamed 6, Bandar Sultan Suleiman, 42000 Port Klang, Malaysia

8. DATE OF REVISION OF THE TEXT

5/12/2024